Readers make targeted regressions to likely word errors in “noisy-channel garden paths”

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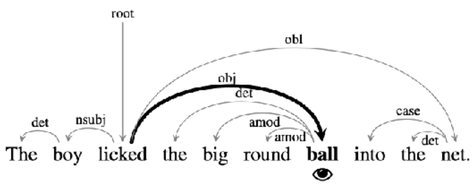
Background

How do readers recover from possible errors in linguistic input during processing?

What is the relationship between regressive eye movements in reading and inferences about alternative sentence interpretations?



Reactivation account: regressions are more likely between pairs of words with high pointwise mutual information

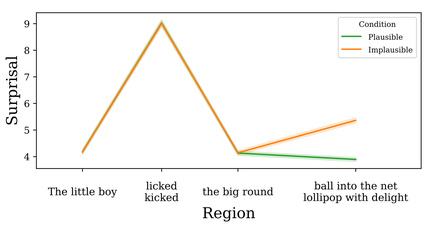
Reanalysis account: regressions are more likely between pairs of words with low pointwise mutual information

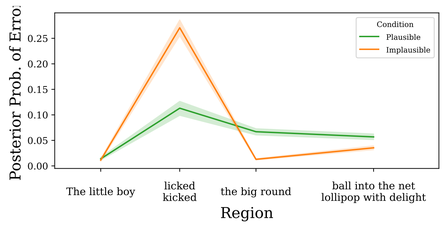
Modeling: Algorithmic models predict non-incremental inference

Surprisal Theory: processing difficulty is a linear function of incremental surprisal

Noisy Channel: processing reflects a *posterior* distribution over alternatives to a possibly noisy input

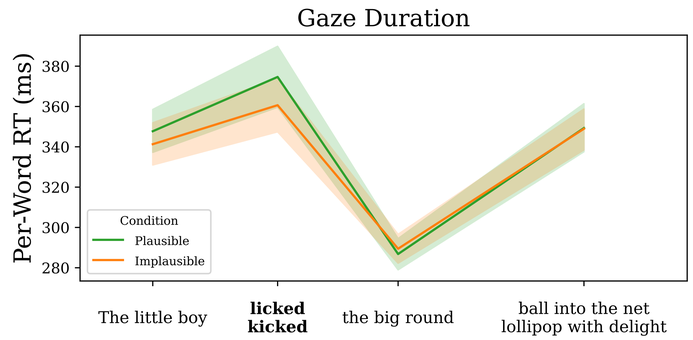
Regions which are incrementally equally surprising may have very different posterior error probabilities [3].

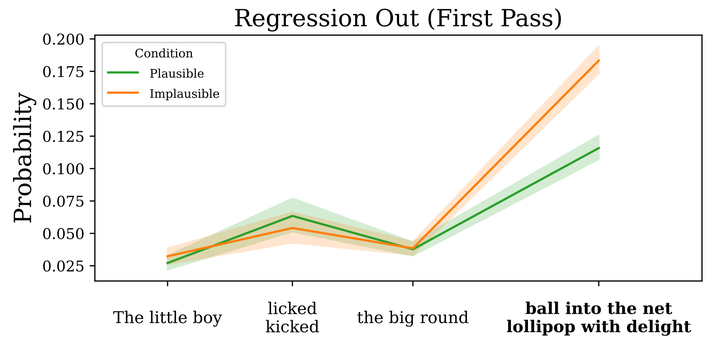


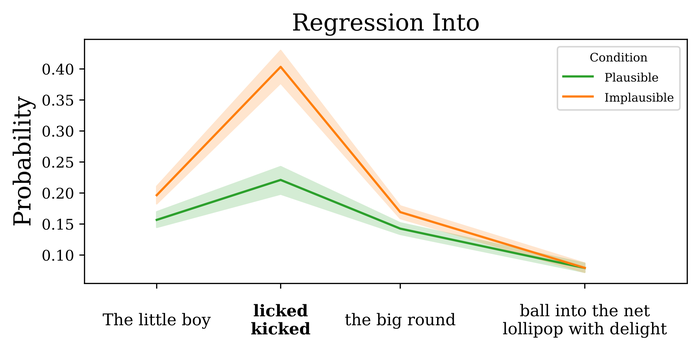


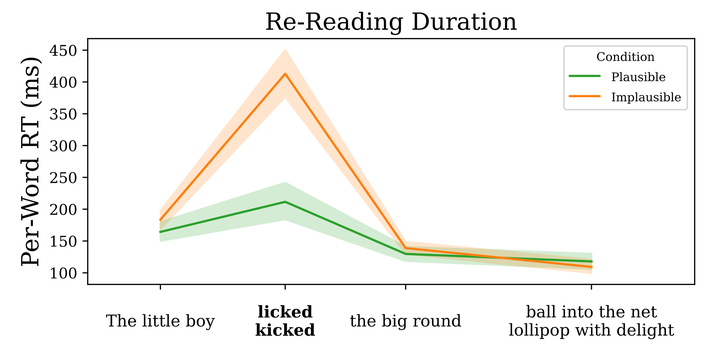
Results

Plots indicate average per-word reading times for each sentence region, with 95% CIs.









We find a significant interaction between Region and Condition for Regressions Out, Regressions Into, and Re-Reading Duration using a Bayesian mixed-effects regression model, with control predictors of surprisal, length, frequency, POS, and word position.

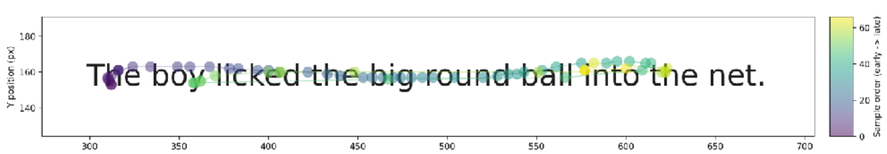
Methods

Alternative	Preamble	Critical Word	Intervening	Predicate
A	the boy	kicked	the big round	ball into the net
A		kicked		lollipop with delight
B		licked		ball into the net
B		licked		lollipop with delight

36 items · N = 200 participants (Prolific) · Mouse Tracking for Reading [8]

Participants judged whether each sentence contained an error (error detection task).

Dependent measures: regression-out probability, regression-into probability, gaze duration, re-reading duration (total minus gaze duration), derived from mouse movements.



Discussion

Key Results:

1. **Regressions out:** more likely out of implausible predicates.
2. **Regressions in** and **re-reading time** are targeted towards locus of likely error (in light of full sentence context).

Takeaways:

- Evidence for **reanalysis account** of regressions, in contrast to the **reactivation account**.
- If re-reading is an option, **readers regress rather than slow down** (no extra reading time on predicate)
- A garden-path-like effect is triggered by **error-sensitive reanalysis**, without the presence of **ambiguity** in the traditional garden-path sense

Additional Analyses and Future Work:

- Explain item-level variation in regression probability
- Test for sensitivity to error probability

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